POWER BI ASSIGNMENT 2 REPORT

Module -2

Power BI Assignment 2 – DAX & Data Visualization

Project Title
E-Commerce Sales Data Analysis using Power BI

SUBMITTED BY
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Batch: ANB18

Contents

Introduction

This assignment focuses on performing data analysis and visualization using Microsoft Power BI on an E-Commerce Sales Dataset. The objective of the project is to apply DAX (Data Analysis Expressions) for creating calculated columns and measures to derive meaningful business insights from the dataset.

The assignment includes analysis of sales amount, profit, targets, order count, and regional performance using interactive dashboards and visualizations. Various chart types such as clustered column charts, line charts, donut charts, scatter charts, treemaps, matrix views, maps, and bar charts were used to present the data in a visually understandable format.

This project helps demonstrate skills in data modeling, DAX calculations, analytical thinking, business reporting, and dashboard design using Power BI.

DAX Calculations

Calculated Columns

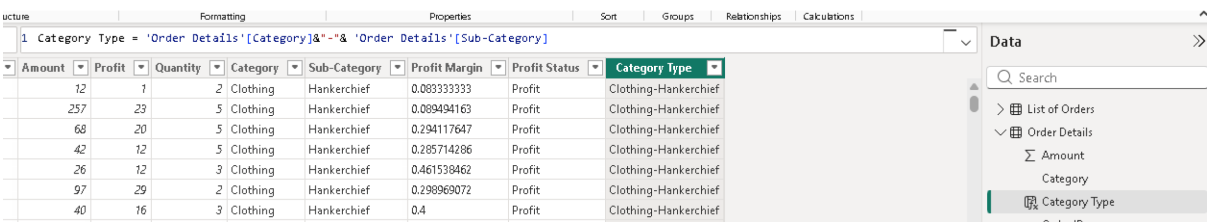
1. **Create a Calculated Column for 'Category Type':** Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

DAX Formula:

Formula Used :Category Type = 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]

Explanation:

This calculated column combines the Category and Sub-Category columns into a single field called "Category Type." It helps provide a more detailed classification of products for analysis and reporting purposes.



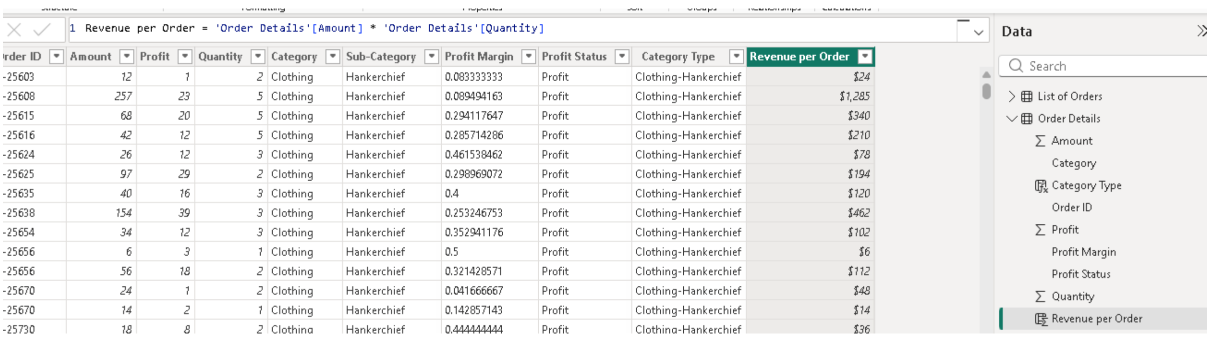
2. **Calculate Revenue per Order in Order Details Table:** Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.

DAX Formula:

Revenue per Order = 'Order Details'[Amount] * 'Order Details'[Quantity]

Explanation:

This calculated column computes the revenue generated for each order by multiplying the sales amount with quantity sold. It helps analyze the revenue contribution of individual orders.



rder ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue per Order
-25603	12	1	2	Clothing	Hankerchief	0.083333333	Profit	Clothing-Hankerchief	$24
-25608	257	23	5	Clothing	Hankerchief	0.089494163	Profit	Clothing-Hankerchief	$1,285
-25615	68	20	5	Clothing	Hankerchief	0.294117647	Profit	Clothing-Hankerchief	$340
-25616	42	12	5	Clothing	Hankerchief	0.285714286	Profit	Clothing-Hankerchief	$210
-25624	26	12	3	Clothing	Hankerchief	0.461538462	Profit	Clothing-Hankerchief	$78
-25625	97	29	2	Clothing	Hankerchief	0.298969072	Profit	Clothing-Hankerchief	$194
-25635	40	16	3	Clothing	Hankerchief	0.4	Profit	Clothing-Hankerchief	$120
-25638	154	39	3	Clothing	Hankerchief	0.253246753	Profit	Clothing-Hankerchief	$462
-25654	34	12	3	Clothing	Hankerchief	0.352941176	Profit	Clothing-Hankerchief	$102
-25656	6	3	1	Clothing	Hankerchief	0.5	Profit	Clothing-Hankerchief	$6
-25656	56	18	2	Clothing	Hankerchief	0.321428571	Profit	Clothing-Hankerchief	$112
-25670	24	1	2	Clothing	Hankerchief	0.041666667	Profit	Clothing-Hankerchief	$48
-25670	14	2	1	Clothing	Hankerchief	0.142857143	Profit	Clothing-Hankerchief	$14
-25730	18	8	2	Clothing	Hankerchief	0.444444444	Profit	Clothing-Hankerchief	$36

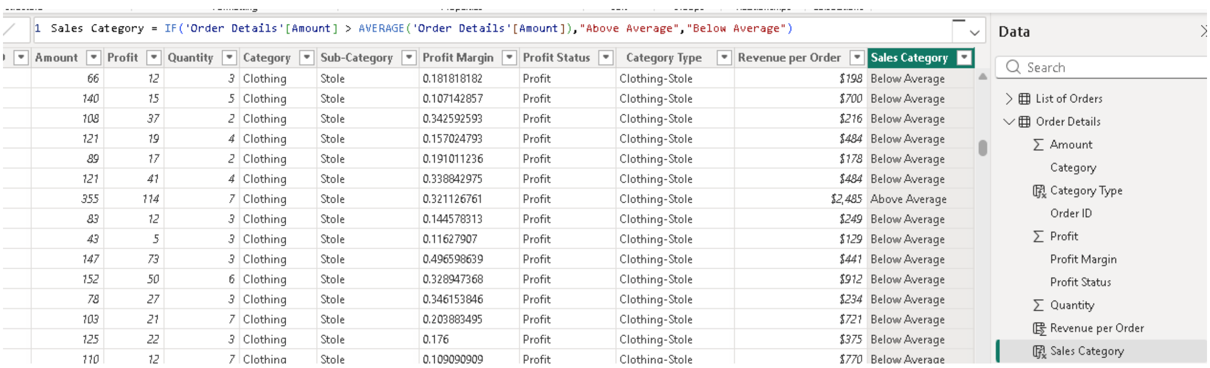
3. **Create a Calculated Column to Categorize Sales:** Add a calculated column named ‘Sales Category’ in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

DAX Formula:

Sales Category =IF('Order Details'[Amount] > AVERAGE('Order Details'[Amount]),"Above Average", "Below Average")

Explanation:

This column categorizes sales transactions into “Above Average” and “Below Average” based on the average sales amount. It helps identify high-performing and low-performing sales records.



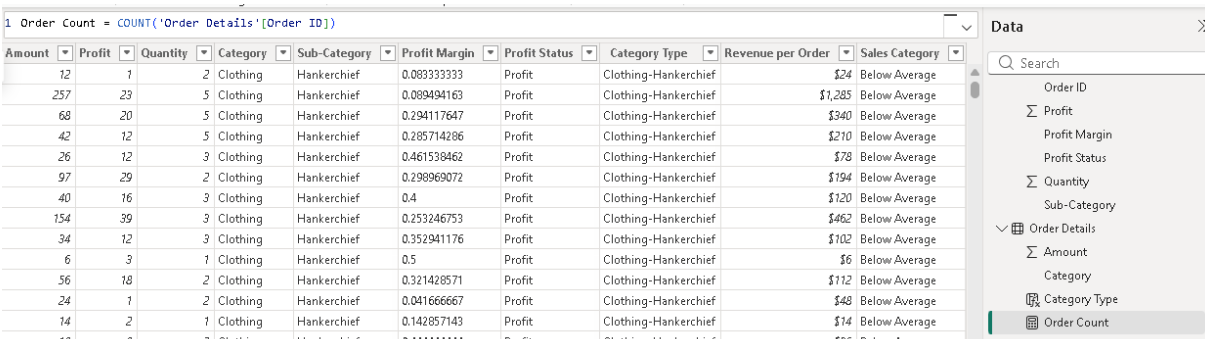
Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue per Order	Sales Category
66	12	3	Clothing	Stole	0.181818182	Profit	Clothing-Stole	$198	Below Average
140	15	5	Clothing	Stole	0.107142857	Profit	Clothing-Stole	$700	Below Average
108	37	2	Clothing	Stole	0.342592593	Profit	Clothing-Stole	$216	Below Average
121	19	4	Clothing	Stole	0.157024793	Profit	Clothing-Stole	$484	Below Average
89	17	2	Clothing	Stole	0.191011236	Profit	Clothing-Stole	$178	Below Average
121	41	4	Clothing	Stole	0.338842975	Profit	Clothing-Stole	$484	Below Average
355	114	7	Clothing	Stole	0.321126761	Profit	Clothing-Stole	$2,485	Above Average
83	12	3	Clothing	Stole	0.144578313	Profit	Clothing-Stole	$249	Below Average
43	5	3	Clothing	Stole	0.11627907	Profit	Clothing-Stole	$129	Below Average
147	73	3	Clothing	Stole	0.496598639	Profit	Clothing-Stole	$441	Below Average
152	50	6	Clothing	Stole	0.328947368	Profit	Clothing-Stole	$912	Below Average
78	27	3	Clothing	Stole	0.346153846	Profit	Clothing-Stole	$234	Below Average
103	21	7	Clothing	Stole	0.203883495	Profit	Clothing-Stole	$721	Below Average
125	22	3	Clothing	Stole	0.176	Profit	Clothing-Stole	$375	Below Average
110	12	7	Clothing	Stole	0.109090909	Profit	Clothing-Stole	$770	Below Average

Calculated Measures

1. **Order Count-** Create a measure that calculates the **total number of orders** in the **Order Details** table.

DAX Formula:
Order Count = COUNT('Order Details'[Order ID])

Explanation:
This measure calculates the total number of orders available in the dataset. It is useful for analyzing overall order distribution and customer purchasing activity.



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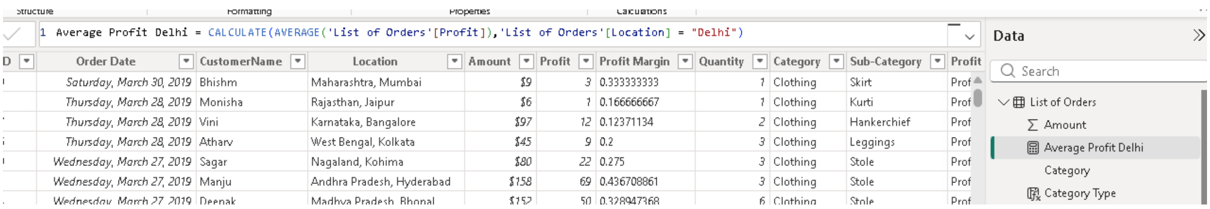
24	1	2	Clothing	Hankerchief	0.041666667	Profit	Clothing-Hankerchief	$48	Below Average
14	2	1	Clothing	Hankerchief	0.142857143	Profit	Clothing-Hankerchief	$14	Below Average

2. **Average Profit in Delhi-** Create a measure that calculates the **average profit for orders placed in Delhi**.

DAX Formula:
Average Profit Delhi =CALCULATE(AVERAGE('List of Orders'[Profit]),'List of Orders'[Location] = "Delhi")

Explanation:
This measure calculates the average profit earned from orders placed in Delhi. It helps analyze city-specific profitability and regional business performance.



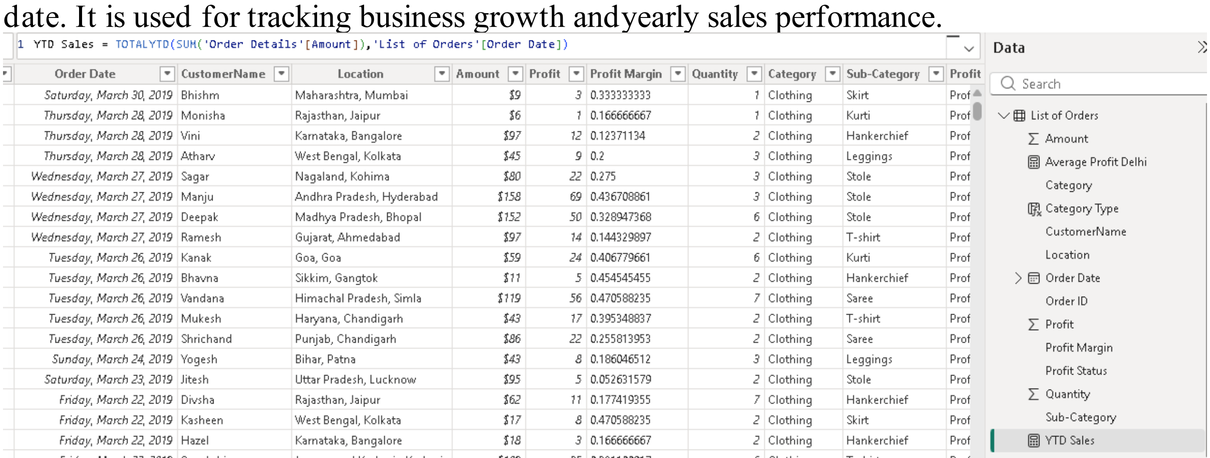
Order Date	CustomerName	Location	Amount	Profit	Profit Margin	Quantity	Category	Sub-Category	Profit
Saturday, March 30, 2019	Bhishm	Maharashtra, Mumbai	$9	3	0.333333333	1	Clothing	Skirt	Prof
Thursday, March 28, 2019	Monisha	Rajasthan, Jaipur	$6	1	0.166666667	1	Clothing	Kurti	Prof
Thursday, March 28, 2019	Vini	Karnataka, Bangalore	$97	12	0.12371134	2	Clothing	Hankerchief	Prof
Thursday, March 28, 2019	Atharv	West Bengal, Kolkata	$45	9	0.2	3	Clothing	Leggings	Prof
Wednesday, March 27, 2019	Sagar	Nagaland, Kohima	$80	22	0.275	3	Clothing	Stole	Prof
Wednesday, March 27, 2019	Manju	Andhra Pradesh, Hyderabad	$158	69	0.436708861	3	Clothing	Stole	Prof
Wednesday, March 27, 2019	Deepak	Madhya Pradesh, Bhopal	$152	50	0.328947368	6	Clothing	Stole	Prof

3. **Year-to-Date (YTD) Sales-** Create a measure that calculates **total sales accumulated from the beginning of the year up to the current order date**

DAX Formula:
YTD Sales = TOTALYTD(SUM('Order Details'[Amount]),'List of Orders'[Order Date])

Explanation:

This measure calculates cumulative sales from the beginning of the year up to the selected date. It is used for tracking business growth andyearly sales performance.



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Wednesday, March 27, 2019	Deepak	Madhya Pradesh, Bhopal	$152	50	0.328947368	6	Clothing	Stole	Prof
Wednesday, March 27, 2019	Ramesh	Gujarat, Ahmedabad	$97	14	0.144329897	2	Clothing	T-shirt	Prof
Tuesday, March 26, 2019	Kanak	Goa, Goa	$59	24	0.406779661	6	Clothing	Kurti	Prof
Tuesday, March 26, 2019	Bhavna	Sikkim, Gangtok	$11	5	0.454545455	2	Clothing	Hankerchief	Prof
Tuesday, March 26, 2019	Vandana	Himachal Pradesh, Simla	$119	56	0.470588235	7	Clothing	Saree	Prof
Tuesday, March 26, 2019	Mukesh	Haryana, Chandigarh	$43	17	0.395348837	2	Clothing	T-shirt	Prof
Tuesday, March 26, 2019	Shrichand	Punjab, Chandigarh	$86	22	0.255813953	2	Clothing	Saree	Prof
Sunday, March 24, 2019	Yogesh	Bihar, Patna	$43	8	0.186046512	3	Clothing	Leggings	Prof
Saturday, March 23, 2019	Jitesh	Uttar Pradesh, Lucknow	$95	5	0.052631579	2	Clothing	Stole	Prof
Friday, March 22, 2019	Divsha	Rajasthan, Jaipur	$62	11	0.177419355	7	Clothing	Hankerchief	Prof
Friday, March 22, 2019	Kasheen	West Bengal, Kolkata	$17	8	0.470588235	2	Clothing	Skirt	Prof
Friday, March 22, 2019	Hazel	Karnataka, Bangalore	$18	3	0.166666667	2	Clothing	Hankerchief	Prof

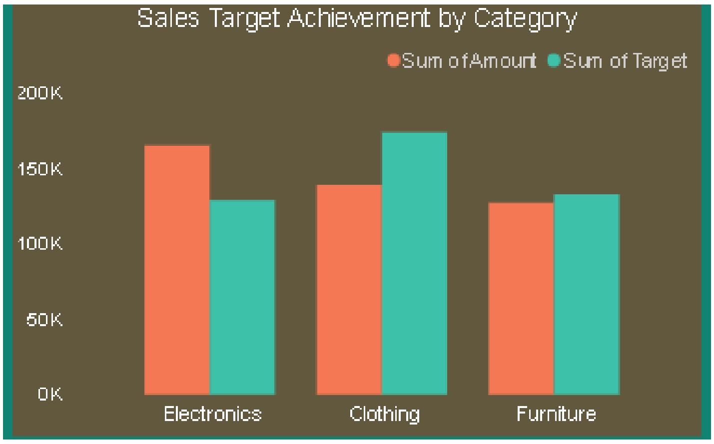
Data Visualization

Create the following visualizations using Power BI:

● **Sales Target Achievement by Category:** Compare actual sales with sales targets by category using a clustered column chart.

Explanation:

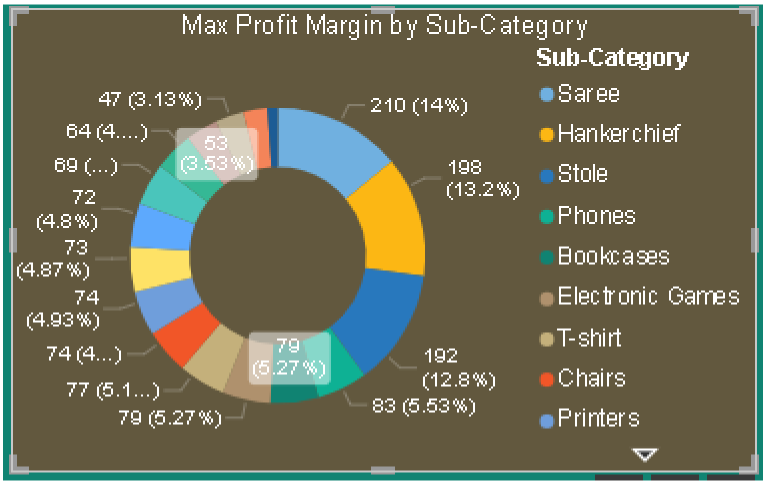
This visualization compares actual sales amount with sales targets across different product categories. It helps identify whether categories are meeting or falling behind business targets.



● **Max Profit Margin by Sub-Category:** Analyze the maximum profit margin for each sub-category of products using a donut chart.

Explanation:

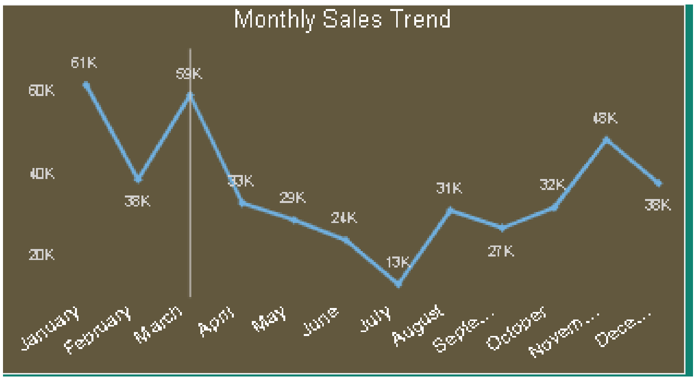
The donut chart displays the profit margin contribution of different product sub-categories. It helps identify which products generate higher profitability for the business.



● **Monthly Sales Trend:** Show the trend of monthly sales over time using a line chart.

Explanation:

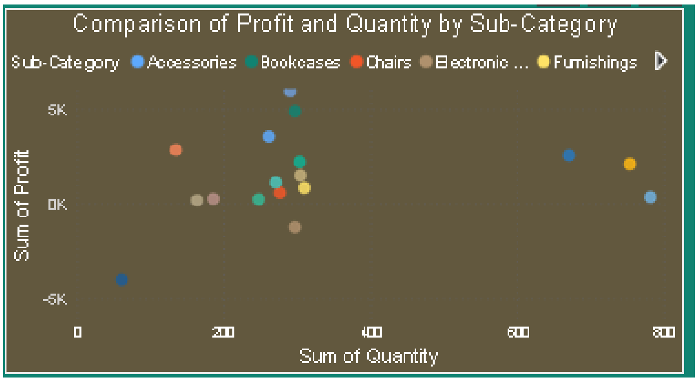
This line chart represents monthly sales trends over time. It helps analyze seasonal patterns, peak sales periods, and business growth trends



● **Comparison of Profit and Quantity by Sub-Category:** Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.

Explanation:

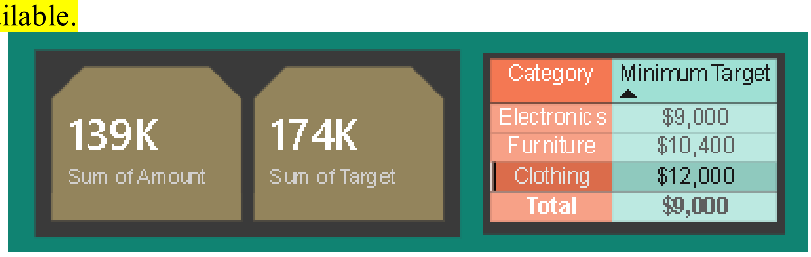
This scatter chart compares profit and quantity sold across product sub-categories. It helps identify products with high sales volume and high profitability.



● **Comparison of Total Sales Amount and Target:** Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.

Explanation:

This card visual displays the total sales amount generated by the business. It provides a quick summary of overall business performance. Using a **Table Visual** instead of Multi-row Card is completely acceptable in Microsoft Power BI, especially if the Multi-row Card option was unavailable.



Sales Performance Matrix: Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.

Explanation:

The matrix visual compares actual sales and target values across categories and months. It helps analyze monthly sales performance and target achievement in a structured format.

Sales Performance Matrix

Month	January		February		March		April		May		June	
Category	Sum of Amount	Sum of Target	Sum of Amount	Sum of Target	Sum of Amount	Sum of Target	Sum of Amount	Sum of Target	Sum of Amount	Sum of Target	Sum of Amount	Sum of Target
Clothing	139054	16000	139054	16000	139054	16000	139054	12000	139054	12000	139054	12000
Electronics	165267	16000	165267	16000	165267	16000	165267	9000	165267	9000	165267	9000
Total	431502	43500	431502	43600	431502	43800	431502	31400	431502	31500	431502	31600

● **Geographic Sales Analysis:** Visualize total sales on a map by city to identify regional sales patterns.

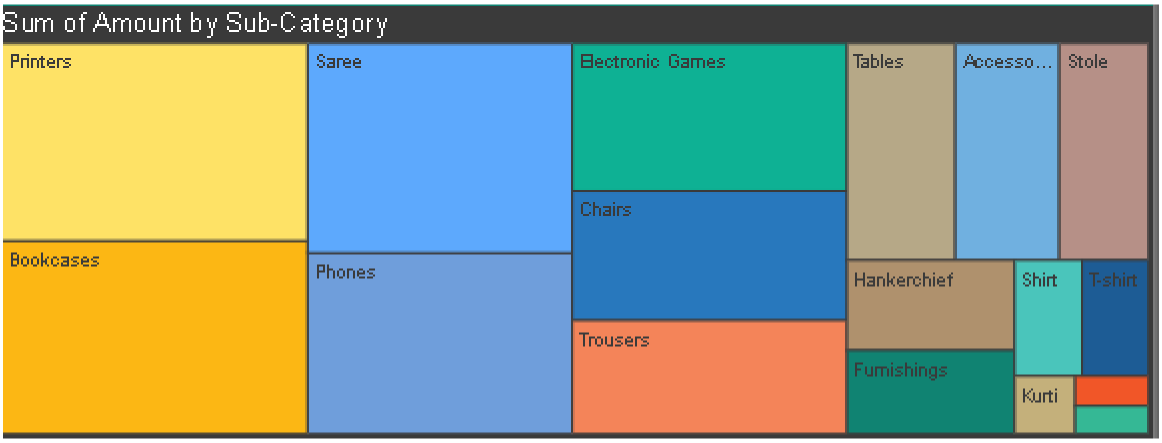
Explanation:

This map visual represents sales distribution across different cities. It helps identify regional sales performance and major revenue-generating locations.

● **Sales Distribution by Sub-Category:** Represent the sales distribution across different sub-categories using a treemap.

Explanation:

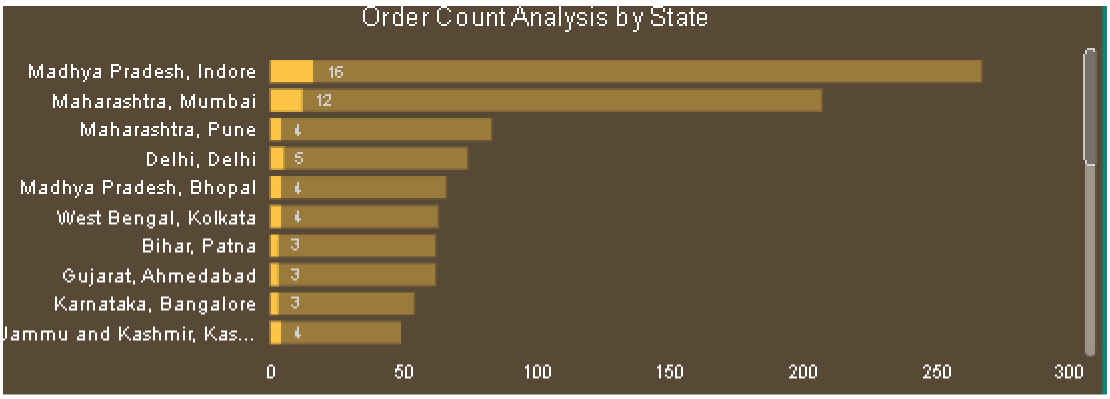
The treemap visual shows the contribution of different sub-categories to total sales. Larger blocks represent higher sales contribution, helping identify top-performing products.



● **Order Count Analysis by State:** Create a funnel chart to visualize the distribution of order counts across different states.

Explanation:

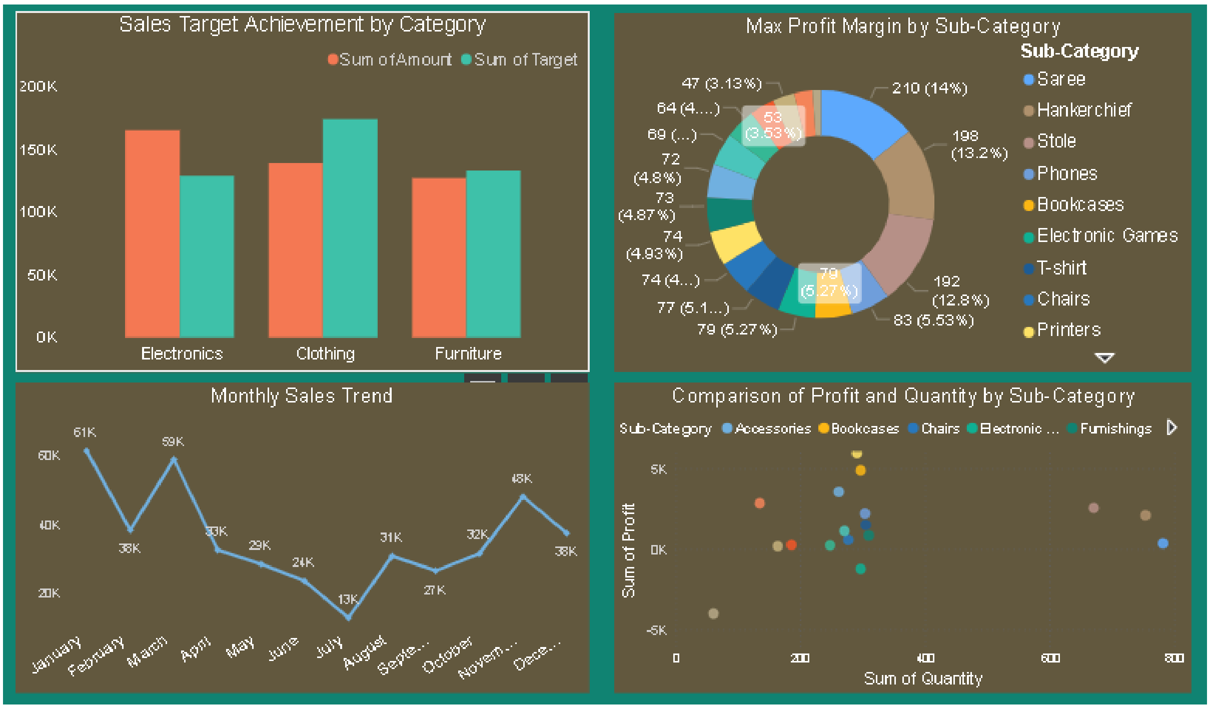
A bar chart was used as an alternative to the funnel chart, as the funnel visual was unavailable in the current Power BI environment. This visualization represents the distribution of order counts across different states and helps identify regions with higher customer order activity and business performance.

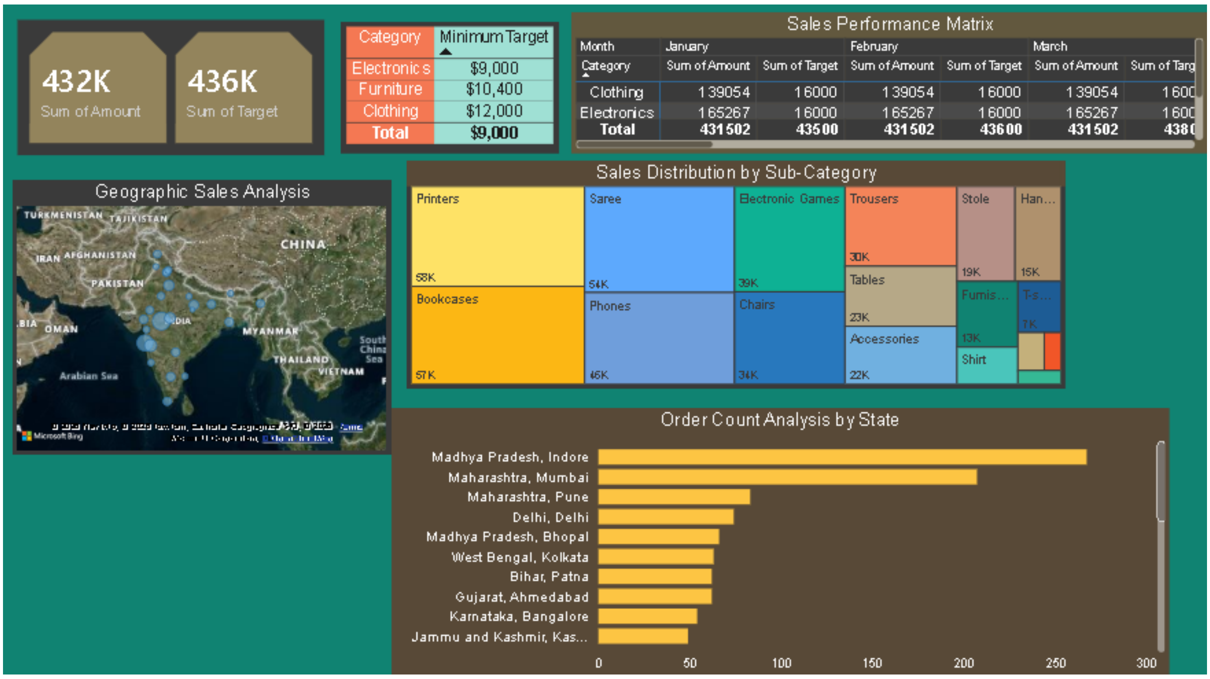


Dashboard Layout

Explanation:

The dashboard combines multiple visualizations into a single interactive reporting interface. It provides insights into sales trends, targets, profitability, regional performance, and product analysis for better business decision-making.





Conclusion

The Power BI dashboard successfully analyzed the E-Commerce Sales Dataset by implementing DAX calculations and multiple visualizations to represent business performance effectively. The report provided insights into sales trends, profitability, category-wise target achievement, regional sales distribution, and customer order analysis.

Through this assignment, important Power BI concepts such as calculated columns, measures, time intelligence functions, data visualization, and dashboard formatting were applied practically. The interactive dashboard enables better understanding of business performance and supports data-driven decision-making.

Overall, this assignment enhanced practical knowledge in Power BI reporting, DAX calculations, and visual storytelling techniques used in real-world business analytics.